

Systematic Literature Review on Comparative Analysis of Stream Ciphers Vs Block Ciphers (with a focus on IoT, mobile and cloud applications)

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I. INTRODUCTION

Cryptographic solutions are now more important than ever in protecting data storage and transmission due to the quick development of connected devices and cloud services. In order to balance performance, energy efficiency, and robust security, mobile, cloud, and Internet of Things (IoT) platforms must overcome numerous obstacles. In these resource-constrained and performance-sensitive environments, cryptographic algorithms—particularly stream and block ciphers—are the foundation of data protection mechanisms [1], [2]. A thorough assessment of stream and block cipher algorithms across cloud, mobile, and Internet of Things platforms is provided by this systematic review of the literature (SLR). The study summarizes the results of 29 publications that give details on the performance of various cryptographic algorithms in diverse real-world situations. Among them are well-known ciphers like AES, ChaCha20, RC4, Blowfish, and many others.

The following research questions serve as a guide for the SLR:

- RQ1. Which stream and block cipher algorithms are most frequently utilized on cloud, mobile, and Internet of Things platforms?
- RQ2. How do these cryptographic methods stack up in terms of security strength and performance (speed, resource usage)?
- RQ3. For each platform, which ciphers offer the optimal trade-off between efficiency and data security?

II. METHODOLOGY

A. Search Strategy

Numerous scholarly databases, including IEEE Xplore, SpringerLink, Elsevier ScienceDirect, ACM Digital Library, MDPI, and Google Scholar, were searched. Research articles on comparative cryptographic performance and security in cloud, mobile, and Internet of Things environments that were published between 2015 and 2025 were included in the search. Among the keywords were:

- "Performance of stream ciphers in cloud/mobile/IoT"
- "Lightweight block cipher for mobile/cloud/IoT"
- "Cryptographic algorithm comparison mobile cloud IoT"

- "Stream Vs Block Cipher"
- "Comparative Analysis between Stream and Block Ciphers"

B. Inclusion Criteria

- Published between 2015 and 2024
- Provides compared evaluation of stream and/or block ciphers
- Experimental results on IoT, mobile, or cloud platforms
- Peer-reviewed and full-text accessible

C. Exclusion Criteria

- Studies introducing only a new cryptographic algorithm without comparative benchmarking
- Focus exclusively on non-IoT embedded hardware / cloud / mobile
- Lacking evaluation or results

D. Data Extraction and Classification

An Excel database was created using the data from 29 completed papers. Every paper was divided into the following categories: The title, author(s), year, journal or conference source, DOI, or citation reference are all included in the basic paper details.

- Platform Type: Each study was labeled according to the platform it was intended for, such as cloud-based systems (e.g., virtual machines, SaaS/PaaS infrastructures), mobile environments (e.g., smartphones, Android systems, BYOD), or Internet of Things devices (e.g., sensor nodes, embedded microcontrollers).
- Cipher Type(s): Determined if the study included both block and stream ciphers, or just one of them.
- Ciphers Evaluated: All cryptographic algorithms examined were documented, including widely used ones like AES, DES, RC4, ChaCha20, Blowfish, RSA, 3DES, ElGamal, and others.
- Security Effectiveness: The performance of the assessed ciphers against established cryptographic techniques was used to summarize each paper.
- Efficiency and Performance: Contains performance metrics like CPU usage, memory footprint, throughput, energy consumption, and encryption/decryption time.

- **Platform Suitability:** Based on results and author conclusions, platform suitability evaluated a cipher's suitability for high-performance applications or environments with limited resources.
- **Complexity of Implementation:** Whether the implementation was scalable, software-optimized, hardware-friendly, or required special configurations was noted.
- **Notable Insights:** The authors' main conclusions, patterns, or suggestions were noted for analysis and report preparation.

E. Summary of Selected Papers

Initially, database searches turned up a total of 50 research articles. Forty-five studies were screened based on their abstracts and titles after five duplicate records were eliminated. 16 of these papers were disqualified because they didn't relate to the review's goals.

The full texts of the final 29 studies were obtained, and their eligibility was evaluated. All 29 were included in the final review since they satisfied the inclusion requirements. The comparative analysis of stream and block ciphers on different platforms was made possible by these papers.

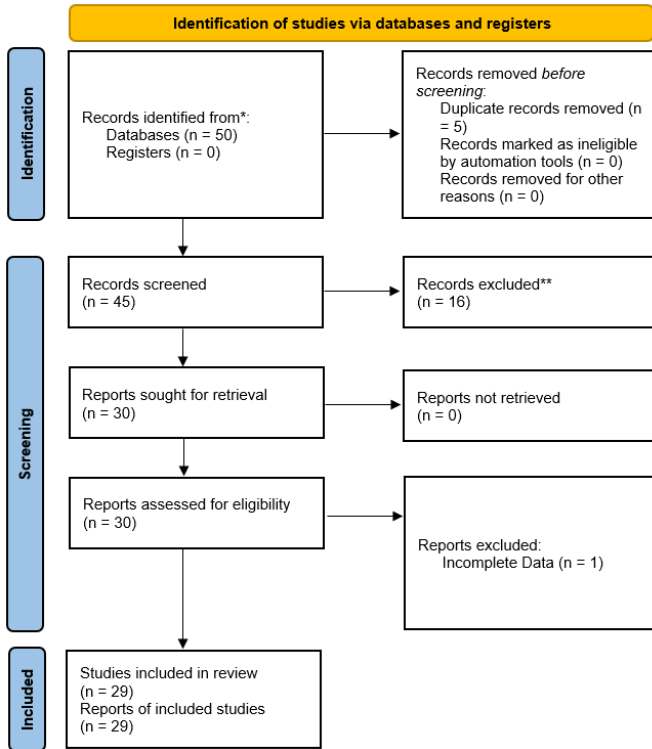


Fig. 1. Prisma Diagram of selected papers

This structured approach ensured a rigorous filtering of studies, retaining only those that were relevant, comparative, and contained concrete cryptographic metric or comparison data.

III. RESULTS

A. Study Characteristics

The performance of stream and block cipher algorithms in three important environments—cloud computing, mobile, and the Internet of Things—is examined and compared in 29 peer-reviewed research papers that are included in this review. Every study was examined based on the kind of platform it was intended for, the cipher types it evaluated, and the publication source. The overall research is broken down as follows: • **Platform Focus:** Of the 29 papers that were chosen, the platform's focus was divided as follows (some papers span multiple platforms); Nine papers looked at mobile platforms, nine papers analyzed cloud-based systems, and eleven papers concentrated on Internet of Things (IoT) environments. **Cipher Types Examined:** Stream and block ciphers were examined in numerous papers, frequently contrasting them in the same context. While popular block ciphers included AES, blowfish, DES, 3DES, Twofish, XTEA, and SPECK, the stream ciphers that were evaluated the most frequently were ChaCha20, RC4, Trivium, Salsa20, and grain. Apart from symmetric algorithms, a number of studies also examined lightweight AEAD schemes like ASCON and TinyJambu, which are especially well-suited for devices with limited resources, and asymmetric encryption techniques like RSA and ElGamal.

Evaluation Metrics: The studies evaluated a number of important criteria in order to compare algorithms. Performance measures how well algorithms resist attacks such as differential, linear, side-channel, and brute-force techniques. These metrics include encryption/decryption time, CPU usage, memory consumption, energy efficiency, and data throughput security. Implementation Complexity, which included aspects like scalability to various systems, ease of key management, and hardware/software suitability.

Sources and databases: In addition to other peer-reviewed journals and conference proceedings, the research papers were taken from reliable academic sources like IEEE Xplore, Springer, Elsevier, ACM Digital Library, Google Scholar, and MDPI. When taken as a whole, these varied studies offer a comprehensive picture of the performance and applicability of cryptographic algorithms today. It provides a strong basis for the comparisons and thematic analysis that follow.

B. Thematic Findings

The performance of cryptographic algorithms in three important environments—the cloud, mobile, and Internet of Things—is the main focus of this section, which also summarizes recurrent themes and insights from the chosen papers. This section summarizes common findings based on platform-specific requirements and limitations rather than going into detail about individual papers.

1) IoT Platforms: The conflict between maintaining efficiency on limited devices and guaranteeing robust security is constantly evident in the assessment of cryptographic algorithms in IoT platforms. Numerous studies examine these issues, providing information on which ciphers work best in particular hardware configurations and operational settings.

The superiority of lightweight ciphers over more conventional algorithms like AES, particularly in extremely constrained nodes, is a significant recurrent theme. Although AES is still the industry standard for security, [3]–[5] points out that its performance overhead is substantial in embedded IoT platforms. Instead, the study assesses block ciphers like PRESENT, RC5, HIGHT, TEA, and XTEA and comes to the conclusion that because of their smaller energy and memory footprints, these are more appropriate for wearables, smart sensors, and medical applications.

This is supported by [6], [7], which benchmarked AES-128, SPECK, and ASCON on Arduino Nano and Micro boards. The results showed that ASCON had a higher processing cost despite being secure, while SPECK provided the highest throughput and lowest latency. This paper also stressed that cryptographic decisions should be platform-driven and that efficiency should not come at the expense of security.

Ciphers like XTEA, Skipjack, and RECTANGLE were found to work best on Raspberry Pi and Arduino Mega platforms in terms of execution time and energy efficiency in a related study, [8], [9]. Despite being extremely secure, AES and PRINCE were found to be less appropriate for low-resource environments because of their greater complexity.

[10], [11] expanded on this hardware focus by assessing the BeagleBone Black and Raspberry Pi 3. In terms of speed and memory efficiency, it was discovered that Twofish (block cipher) and ChaCha20 (stream cipher) performed better than the others, particularly on the Raspberry Pi 3. It underlined that stream ciphers are perfect for real-time or battery-sensitive Internet of Things use cases because they typically provide faster and easier implementations.

[12], [13] also focused on platform-specific performance comparisons. The authors found that AES-CTR, Trivium, and LetterSoup provided the best performance when weighing speed, energy consumption, and platform compatibility using TelosB motes and Intel Edison. The study also emphasized that developers must take into account both hardware and software stacks when choosing a cipher, and that results varied depending on OS-level optimizations (TinyOS vs. ContikiOS).

[14] provides a more comprehensive algorithmic comparison, concentrating on 32 NIST LWC Round 2 AEAD candidates. It identified ciphers like SPARKLE, TinyJambu, and Saturnin as exceptional performers and classified many algorithms as efficient for software-based embedded platforms, without going into detail about the performance of each algorithm. It also confirmed previous findings by pointing out that AES-based AEADs (such as AES-GCM) were typically less appropriate for limited IoT systems.

The trade-offs between symmetric key algorithms in limited IoT networks were investigated in [15], [16]. According to the study, algorithms like 3DES and RC5 are no longer practical because of their high computational cost and antiquated security, whereas AES and Blowfish offer a strong balance between security and performance. According to the paper, Blowfish provides a good substitute for lower-power nodes, while AES can be appropriate when used with hardware acceleration

or in devices with moderate processing power. The authors emphasized the necessity of modular implementation across layered IoT communication systems and compatibility with lightweight protocol stacks.

52 block ciphers were analyzed using contemporary cryptanalysis techniques in security-focused reviews like [17], [18]. It came to the conclusion that while SPECK, Piccolo, and PRIDE provide a better balance between implementation cost and security, PRESENT, CLEFIA, and AES are among the most secure and thoroughly investigated options. Crucially, this paper presented the notion that SPN structures are generally more secure than Feistel networks, which affects the choice of cipher in the design of IoT protocols.

Additionally, [19], [20] focused on stream ciphers. After evaluating several ciphers on the ATmega328P, including Lizard, Plantlet, Sprout, Grain v1, and Fruit, the authors concluded that Lizard, Sprout, and Plantlet offered the best security, power, and speed balance. These stream ciphers were also commended for their small hardware footprints, which made them ideal for microcontrollers with limited resources, and their resilience to statistical tests.

AES provides universal applicability, but ciphers like SPECK and PRESENT are far more appropriate when resource constraints are crucial, according to the study [21], which also introduced a unique comparison framework. Additionally, it noted that algorithm selection is practically influenced by developer familiarity and available cryptographic libraries.

In addition to standard cipher analysis, [22] suggested using Rijndael S-box and mutual authentication in conjunction with a unique lightweight block cipher. The study, which concentrated on power savings and fewer CPU cycles in smart healthcare IoT systems, found that, when designed with the application's domain in mind, custom lightweight protocols can occasionally perform better than conventional ones.

a) Conclusion of Thematic Synthesis for IoT devices:

According to the combined results of these eleven papers, there isn't a single cryptographic algorithm that is always the best for the Internet of Things. Rather, platform-level limitations, application domain, and hardware architecture all have a significant impact on efficacy. AES is still the industry standard for security, but ciphers like SPECK, XTEA, ChaCha20, Trivium, and Lizard repeatedly show up as promising options for the Internet of Things because of their low energy consumption, straightforward design, and reliable performance on limited devices. It is advised that researchers and developers match the choice of cipher to particular deployment scenarios and stay up to date on the latest advancements in cryptanalytics pertaining to lightweight cryptography standards.

2) Mobile Platform: The deployment of cryptographic algorithms is subject to special limitations on mobile platforms, such as limited memory and battery life, varying hardware acceleration support, and user-interactive performance requirements. Together, the chosen papers highlight these limitations and assess the effectiveness of both traditional and lightweight encryption methods in actual mobile settings.

The study [23], [24] emphasize how crucial it is to maximize cryptographic performance at various device tiers. For example, the Galaxy S9+ and A20e smartphones were used to test AES, DES, 3DES, RC4, ChaCha20, and Blowfish. ChaCha20 was the fastest on the more potent S9+, while RC4 beat other algorithms on the less powerful A20e, demonstrating how hardware capability is a key factor in performance results.

[25] further supported ChaCha20's superior performance by demonstrating that, on mobile devices without AES hardware support, ChaCha20-Poly1305 was almost 3× faster than AES-GCM. Because of its low latency and CPU load, the study suggests ChaCha20 as a preferred substitute in mobile TLS environments [25].

In [26], hardware-secured mobile environments were investigated. Using Android phones backed by TEE and RPMB, ChaCha20-Poly1305 was contrasted with AES-XTS. The findings demonstrated ChaCha20's benefit even in security-enhanced scenarios, showing 2.36× faster reads and 1.27× faster writes [26], [27].

Algorithm suitability across different platforms and access levels was also investigated in extensive comparative studies. In [28], ARMv7-a and ARMv8-a smartphones were used to test ciphers such as ChaCha20-Poly1305, SPECK, LEA, and AES. SPECK and LEA were the most battery-efficient, particularly in basic-access use cases, while ChaCha20-Poly1305 offered the highest throughput [28].

Similarly, AES, RC6, Twofish, and SPECK were evaluated by [21]. The results showed that AES remained dominant when hardware acceleration was available, while Twofish and SPECK provided a better performance-security trade-off on devices that were limited or mid-range.

Energy optimization and execution time were the main topics of several papers. For instance, AES, DES, TEA, and a custom method (REA) were compared in [29], which found that AES consistently showed the lowest execution time and memory consumption. AES, Blowfish, Twofish, RC4, and 3DES were all assessed in [30], with RC4, Twofish, and 3DES demonstrating the optimal speed-to-resource balance for real-time mobile applications.

Evaluations that focused on stream ciphers were covered in [22], wherein ARM-based Android phones were used to test Salsa20, Trivium, Grain-128a, and Sosemanuk. Because of their high throughput and low CPU overhead, these stream ciphers demonstrated a strong suitability for real-time sensor-based apps, live video, and messaging.

One study, [31], presented a specially designed block cipher for Android that makes use of Rijndael S-boxes and CBC mode. With the goal of improving mutual authentication and encryption in mobile communication, it reported remarkably low power and CPU usage [31].

AES, Blowfish, RSA, and ElGamal were evaluated in a BYOD environment in another paper with a real-world enterprise focus, [32]. When it came to integrating Java-based cryptographic libraries for employee-connected smartphones and tablets, AES and Blowfish were the quickest and most straightforward [32].

Finally, [33] looked at algorithms such as Blowfish and AES in both mobile and IoT contexts. Its results [30], [34] supported AES and Blowfish as quick and portable options for mobile applications when secure key handling is necessary, even though they weren't specifically mobile-centric.

a) Conclusion of Thematic Synthesis for Mobile Devices: In conclusion, the reviewed studies highlight how implementation environments, energy constraints, and device hardware all have a significant impact on mobile cryptographic performance. While AES remained competitive where hardware acceleration was available, stream ciphers such as ChaCha20 and RC4 continuously outperformed conventional block ciphers on lower-end devices. Lightweight substitutes like LEA, Twofish, and SPECK worked well to reduce resource consumption and battery drain. The function of context-specific cryptography in mobile environments was further demonstrated by enterprise-specific deployments and customized solutions. Overall, the results validate an increasing trend in the security of contemporary mobile applications toward algorithms that are energy-efficient, hardware-independent, and lightweight.

3) Cloud Platform: [35] In this paper, ChaCha20 is proposed as a stream cipher solution for cloud data center security, especially when combined with Merkle hash tree and SHA-256 techniques. It showed that ChaCha20 offers improved resistance to timing and side-channel attacks and executes encryption tasks roughly three times faster than AES-128. It was discovered that the cipher's 256-bit key strength and constant-time execution made it ideal for virtualized environments where multi-tenant data handling security and speed are crucial. [36], [37] Weighted attribute-based encryption (BH-WABE) and symmetric Blowfish encryption were combined in this study to create a hybrid model. The plan sought to balance performance, scalability, and confidentiality while offering fine-grained access control in collaborative cloud environments. In contrast to other attribute-based models like HABE, the authors found that BH-WABE greatly decreased decryption time and complexity, making it ideal for data sharing between several authorities in cloud systems.

[38], [39] This thorough analysis evaluated the use of conventional cryptographic methods in cloud infrastructures, including AES, 3DES, Blowfish, RSA, and homomorphic encryption. While RSA and Homomorphic Encryption provided stronger security at the expense of computation overhead, AES was found to consistently achieve better performance and shorter execution times. The trade-off between processing speed and security strength was highlighted in the paper, which supported the applicability of Blowfish and AES in cloud applications where encryption speed and response time are crucial.

[40] AES, DES, and Blowfish were compared in this paper along with a suggested encryption method that was put into practice using MATLAB. The authors noted that although conventional block ciphers are still useful, creative solutions are required due to their shortcomings in handling cloud-specific risks like insider attacks and data breaches.

Although not entirely comprehensive, the suggested approach demonstrated improved adaptability and performance for cloud use cases involving secure access control and real-time data protection.

[41] A lightweight encryption algorithm created especially for cloud platforms, NLCA, was proposed in the paper. When compared to DES, AES, Blowfish, HIGHT, and LED, NLCA performed better in terms of encryption speed and avalanche behavior. The authors highlighted how NLCA is ideal for scalable or limited cloud systems due to its low implementation complexity and capacity to avoid resource-intensive operations like large multiplications.

[42], [43] This paper compared Hyper-V and KVM hypervisors using cryptographic algorithms like AES, RSA, Blowfish, TripleDES, Twofish, and CAST-128 with an emphasis on the virtualization layer of cloud platforms. The findings showed that KVM performed better than Hyper-V in every metric that was assessed, especially in terms of encryption throughput and system resource efficiency. For cloud deployments to operate at their best, these insights emphasize how crucial it is to choose the appropriate virtualization infrastructure in addition to cryptographic algorithms.

[44] Another study that evaluated NLCA reaffirmed the algorithm's strength in terms of avalanche effect, simplicity of operations like XOR/XNOR, and execution time (specifically 1.18s for 128×128 image encryption). NLCA provided more randomness and better performance than AES, DES, and Blowfish, particularly in cloud-based image/file protection scenarios where computational efficiency is crucial.

[45] In this paper, multi-factor access control schemes for cloud environments were introduced along with the Functional-Based Stream Cipher (FBSC). The model, which was created for situations that needed data access by multiple users and authorities, showed enhanced encryption times and security using methods like Lagrange interpolation and Shamir's Secret Sharing. The cipher supported secure cloud services without introducing undue computational overhead by successfully striking a balance between performance and access flexibility.

[46] Blowfish was the fastest cipher in this performance-based study, followed by AES, while 3DES and RSA trailed behind. High-end Intel hardware running Python was used for the tests, which replicated actual cloud storage scenarios. The outcomes demonstrated Blowfish's superior encryption/decryption speed, confirming its applicability for quick cloud storage encryption, especially in settings that demand quick data retrieval and classification.

a) Conclusion of Thematic Synthesis for Cloud Platform: Thematic synthesis of cloud-focused research shows a steady trend toward efficient, lightweight, and hybrid cryptographic methods that meet the unique requirements of cloud environments, including low latency, virtual machine scalability, and multi-user access.

Newer techniques like ChaCha20, NLCA, and hybrid schemes (like BH-WABE) show great promise in enhancing encryption speed, resource efficiency, and security flexibility,

even though AES and Blowfish are still popular and efficient. These results emphasize how crucial it is to modify cryptographic solutions to fit the changing architecture and performance requirements of contemporary cloud platforms.

Future cryptographic designs will probably place a higher priority on side-channel attack resistance, strong access control, and low-complexity operations as cloud applications continue to grow. This will guarantee both performance and reliability in dynamic, distributed environments.

C. Statistical Synthesis

This statistical synthesis excluded some of the algorithms that were discussed in the Thematic Synthesis, for e.g. SPARKLE, Saturnin, TinyJambu. This is due to the fact that they were only assessed in limited or one-time experimental setups or lacked comprehensive and consistent performance metrics across studies. Therefore, algorithms that have been repeatedly benchmarked using standardized metrics across various platforms are the focus of the statistical synthesis.

1) IoT Platforms: The thematic finding that simplified, light-weight ciphers perform noticeably better than conventional algorithms in limited settings is supported by quantitative performance data from IoT platforms. TelosB motes, Intel Edison, Raspberry Pi 3, and Arduino microcontrollers are among the hardware platforms covered in the analysis; each shows unique trends according to processing power and operational limitations.

[12] [10] [6] [8] With an init time of 0.47 ms and an update time of 0.91 ms, AES-CTR was determined to be the fastest encryption algorithm on TelosB devices by. Blake2s and Marvin both showed excellent performance for hashing and MACs; Blake2s finished hashing in 11.94 ms (128B), while Marvin finished MACs in 7.04 ms. With an init time of 3.04 ms and the lowest reported energy use among tested AEAD techniques, LetterSoup performed best for AEAD operations on TelosB.

The same study demonstrated that, with hardware-level optimizations and software integration through Yocto OS, AES performed better than alternatives on Intel Edison for payloads up to 32 bytes, with initial and final times below 7.33 μ s.

ChaCha20 was found to be the most effective stream cipher on the Raspberry Pi 3, surpassing AES, DES, and 3DES in terms of execution time for big payloads, according to [Performance Evaluation of Cryptographic Ciphers on IoT Devices]. In contrast, RECTANGLE performed exceptionally well for message sizes up to 2048 bytes, while XTEA was the best performer for smaller message sizes (16–128 bytes) in a payload-specific evaluation, suggesting that cipher efficiency may scale differently depending on data size.

According to the studies, SPECK 128 offered the best combination of speed (273 μ s), encryption throughput (824 Kbps), and low RAM/ROM usage (1164B / 8848B) for Arduino-class microcontrollers. Despite having a stronger cryptography, ASCON had the highest latency (849 μ s), which makes it more suitable for high-security IoT use cases where throughput is not as important.

Platform	Category	Best Algorithm	Init Time	Update/Encrypt Time	Final Time	Energy Use
TelosB (TinyOS)	Encryption (AES-CTR)	AES-CTR	0.47 ms	0.91 ms	0.91 ms	Low (proportional to time)
TelosB (TinyOS)	Hash (Blake2s)	Blake2s	0.96 ms	11.94 ms (128B)	6.27 ms	70 uJ (128B)
TelosB (TinyOS)	MAC (Marvin)	Marvin	5.62 ms	14.50 ms (108B)	7.04 ms	Not specified
TelosB (TinyOS)	AEAD (LetterSoup)	LetterSoup	3.04 ms	23.96 ms (108B w/ AAD)	Not specified	Low (best among AEADs)
Intel Edison (Yocto)	Encryption (AES-128)	AES	7.33 us	<= 5 us (<=32B)	<= 5 us	0.49 J (AES)
Intel Edison (Yocto)	Encryption (Twineam-32B)	Twineam	11.32 us	<= 4 us (<=32B)	<= 4 us	0.89 J (Twineam)
Intel Edison (Yocto)	Hash (Blake2s)	Blake2s	5.95 us	12 us (128B)	24.09 us	0.12 J (Blake2s)
Intel Edison (Yocto)	MAC (HMAC)	HMAC	32.90 us	15 us (128B)	39.15 us	7.06 J

Fig. 2. IoT Performance Metrics - Telos B and Intel Edison

Platform	Category	Best Algorithm	Encrypt Time
Raspberry Pi 3	Block Cipher (ECB)	Twofish	63.80815 s
Raspberry Pi 3	Block Cipher (CBC)	Twofish	61.11897 s
Raspberry Pi 3	Stream Cipher	ChaCha20	57.66328 s
Raspberry Pi 3	Stream Cipher	RC4	61.88362 s
BeagleBone Black	Block Cipher (ECB)	Twofish	110.3987 s
BeagleBone Black	Block Cipher (CBC)	Twofish	114.9821 s
BeagleBone Black	Stream Cipher	ChaCha20	109.138972 s
BeagleBone Black	Stream Cipher	RC4	112.001322 s

Fig. 3. IoT Performance Metrics - Raspberry Pi3 and Beaglebone Black

Platform	Category	Best Algorithm	Init Time	Update/Encrypt Time	Final Time	Energy Use
Arduino Nano	LWC	SPECK 128	273.73 us	824041 bit/s	793214 bit/s	RAM: 1164 ROM: 8848
Arduino Nano / Micro	Block Cipher	AES	94.93 us	226041 bit/s	118601 bit/s	RAM: 1570 ROM: 10928
Arduino Nano / Micro	Block Cipher	SPECK 128	273.73 us	824041 bit/s	793214 bit/s	RAM: 1164 ROM: 8848
Arduino Nano / Micro	Block Cipher	ASCON	849.67 us	188349 bit/s	187516 bit/s	RAM: 940 ROM: 1108

Fig. 4. IoT Performance Metrics - Arduino Nano/ Micro

According to the studies, SPECK 128 offered the best combination of speed (273 μ s), encryption throughput (824 Kbps), and low RAM/ROM usage (1164B / 8848B) for Arduino-class microcontrollers. Despite having a stronger cryptography, ASCON had the highest latency (849 μ s), which makes it more suitable for high-security IoT use cases where throughput is not as important.

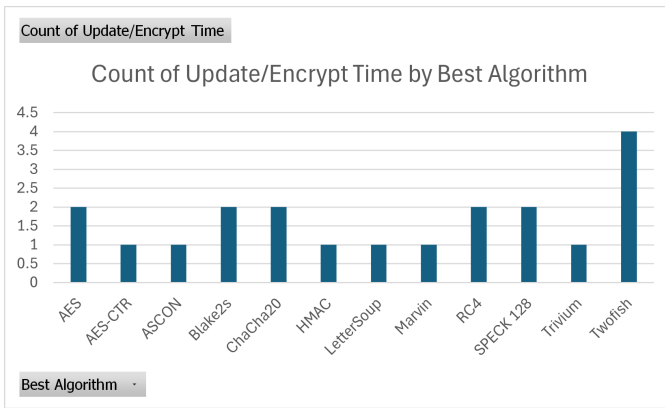


Fig. 5. IoT - Count of Update-Encrypt Time by Best Algorithm

No single cryptographic algorithm consistently performs better than others across all metrics and platforms in IoT environments, according to the quantitative analysis. Rather, the hardware capabilities and operational environment of the platform on which an algorithm is implemented have a significant impact on its efficiency, energy consumption, and memory usage.

Algorithms like SPECK, ChaCha20, XTEA, and AES-CTR have continuously shown themselves to be the best; SPECK is faster and more memory-efficient on devices with limited resources, like Arduino Nano, while AES-CTR works best on platforms with OS-level optimization, like Intel Edison. The trade-off between security and performance was highlighted by ASCON's higher latency despite its stronger cryptography.

Crucially, the statistical synthesis confirms findings from the qualitative analysis: in IoT systems, lightweight cryptographic algorithms are essential for preserving security without sacrificing responsiveness or energy efficiency. The findings are consistent with a context-driven cipher selection approach, in which developers and system architects must assess cryptographic options based on real-world performance benchmarks that are specific to their communication requirements and device limitations in addition to theoretical strength. [12] [10] [6] [8]

2) **Mobile Platform:** [31] [26] [19] A small number of studies that offered comprehensive and repeatable performance metrics, such as encryption throughput, execution time, battery consumption, and memory usage, served as the foundation for the quantitative analysis of mobile cryptographic performance. Only specific algorithms were measured due to a lack of data, so this section only looks at ciphers that were used repeatedly in several studies with distinct benchmarks for comparison. Due to limited or non-standardized data, some of the algorithms covered thematically in Section 3.2, including REA, RSA, ElGamal, and some custom or stream ciphers, are not included here.

Cipher	Device	Exec Time (ms)
AES-128 (GCM)	Galaxy S9+	1629.124
AES-256 (GCM)	Galaxy S9+	1745.916
AES-128 (GCM)	Galaxy A20e	3867.815
AES-256 (GCM)	Galaxy A20e	3845.004
RC4	Galaxy S9+	682.013

Fig. 6. Mobile Performance Metrics - Execution time

Cipher	Device	Exec Time (ms)	Mem Usage (MB)
RSA	Generic Android	860	17.8
Blowfish	Generic Android	360	5
AES	Generic Android	390	5
3DES	Generic Android	430	5

Fig. 7. Mobile Performance Metrics - Execution time and memory usage

On actual Android devices, ChaCha20 and RC4 consistently demonstrated superior performance among the assessed algorithms. This shows that cipher performance scales significantly with hardware capability. In the studies, RC4 had the shortest execution time on low-end devices like the Galaxy A20e, while ChaCha20 beat all others on higher-end smartphones like the Galaxy S9+.

ChaCha20-Poly1305 continuously produced throughput above 130 MiB/s on ARMv8-a Xiaomi devices and 38.95 MiB/s on Samsung devices, according to a thorough evaluation

Cipher	Device	Enc Throughput (MiB/s)	Battery Usage (mAh)
AES-128-GCM	Samsung	12.989	13.4
AES-128-GCM	Xiaomi	77.586	3.14
RC6-128-GCM	Samsung	17.328	8.68
RC6-128-GCM	Xiaomi	89.785	2.38
SPECK128	Samsung	23.848	5.18
SPECK128	Xiaomi	94.345	1.98
LEA	Samsung	19.454	6.65
LEA	Xiaomi	91.236	1.99

Fig. 8. Mobile Performance Metrics - Encryption throughput and Battery Usage

Cipher	Device	Enc Throughput (MiB/s)
ChaCha20-Poly1305	Samsung	38.95
ChaCha20-Poly1305	Xiaomi	138.336

Fig. 9. Mobile Performance Metrics - Encryption throughput - Chacha20-Poly1305

of throughput and battery efficiency in the studies. AES-GCM, on the other hand, achieved slower speeds, particularly in software-based scenarios without hardware acceleration. Furthermore, the least amount of battery was used by lightweight ciphers such as SPECK128 and LEA, which on Samsung devices registered as low as 5.18 and 6.65 mAh, respectively.

Comparisons of execution time and memory usage from studies showed that 3DES, RC4, and Twofish performed well in terms of execution time and memory usage, with Twofish being the most effective block cipher examined. AES and Blowfish also showed moderate encryption speed and low memory requirements (as low as 5 MB) in the studies.

RSA and other asymmetric schemes, on the other hand, required up to 17.8 MB of memory and 860 ms for a full encryption operation, despite being secure. This encourages their continued use on mobile endpoints for authentication or key exchange instead of complete data encryption.

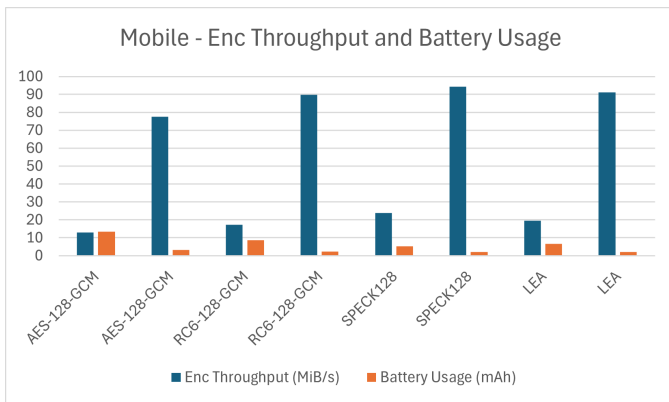


Fig. 10. Mobile Performance Metrics - Battery Usage and Enc. Throughput graph

The quantitative synthesis concludes that ChaCha20, RC4, Twofish, and AES (with hardware acceleration) are effective and sensible options for contemporary mobile applications. The selection of algorithms is still based on the trade-off be-

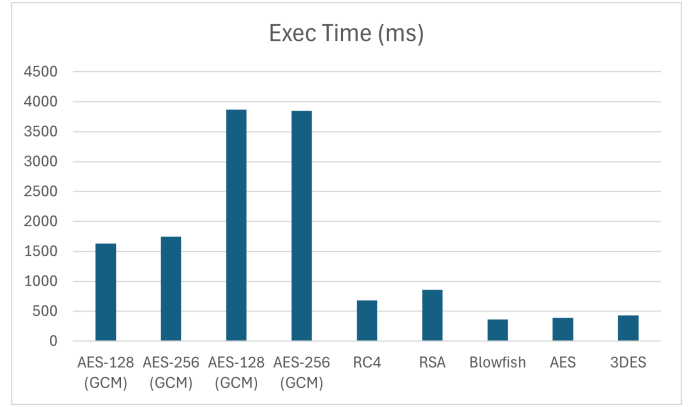


Fig. 11. Mobile Performance Metrics - Execution time

tween speed, battery life, and memory footprint, underscoring the importance of aligning cryptographic designs with device capabilities and use cases. [31] [26] [19]

3) **Cloud Platform:** [38] [41] [42] In order to enable more thorough cross-dimensional comparison, this synthesis focuses on algorithms with a variety of performance metrics. Entropy, CPU usage, avalanche effect, encryption/decryption time, RAM usage, and execution time are among the metrics examined.

Algorithm	Platform	Enc Time (ms)	Dec Time (Automated) (ms)	CPU Usage(%)
RSA	KVM	20.5	30.5	70.2 / 68.1
RSA	Hyper-V	24.2	38.1	73.6 / 71.7
AES	KVM	3.1	3	41.0 / 42.6
AES	Hyper-V	3.5	3.5	45.4 / 47.4
TripleDES	KVM	11.6	11.7	44.9 / 46.4
TripleDES	Hyper-V	11.9	12.1	48.2 / 49.1

Fig. 12. Cloud Platform Performance Metrics - KVM, Hyper - V platforms

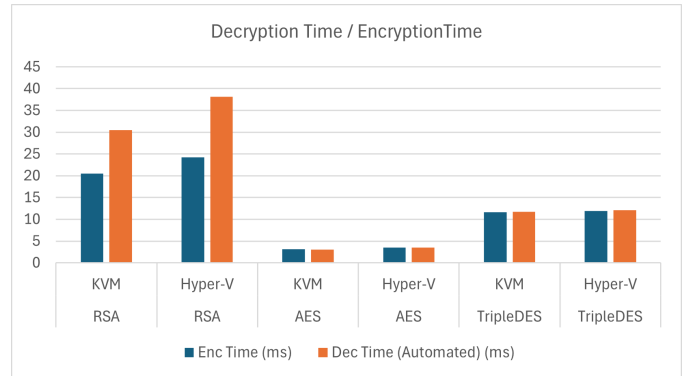


Fig. 13. Cloud Platform Performance Metrics - Decryption Time / Encryption Time

AES continuously produced outstanding outcomes in a variety of performance areas. With 3.1 ms (KVM) and 3.5 ms (Hyper-V), it showed the fastest encryption and decryption times. Even across hypervisors, decryption maintained an efficiency of 3.0 to 3.5 ms.

AES does not overload virtualized cloud systems, as evidenced by the moderate CPU usage, which ranged from 41 percent to 47 percent. AES demonstrated strong resistance to differential attacks by maintaining strong avalanche effects in terms of diffusion (0.50 for key changes and 0.55 for plaintext).

AES was perfect for cloud platforms that need speed and a small footprint because of its low memory usage of only 14.7 KB. These outcomes support AES's standing as the most well-rounded and cloud-optimized block cipher in practical applications.

Another test revealed a higher key-based avalanche of 0.6756, indicating decent diffusion properties, while DES demonstrated respectable avalanche scores of 0.53 for both key and plaintext.

Its vulnerability and inefficiency by today's standards are highlighted by its brute-force simulation decryption time of $2.0 * 10$ hours. DES is not appropriate for production cloud environments, despite being educational.

SF did well on a number of axes. It came very close to ideal randomness with a high entropy (7.99) and a respectable avalanche score of 0.457. Its low RAM usage of 22 KB makes it ideal for light-duty applications.

Although its 2.110-second execution time is marginally slower than that of more efficient ciphers, it is still suitable for cloud services with limited resources where memory footprint is a top concern. For cloud applications with moderate security and low resource requirements, SF is a good choice.

Among lightweight ciphers, HIGHT had the fastest execution time (1.615 seconds) and the highest entropy (7.9971).

Considering its performance, this trade-off is acceptable even though it required more RAM (288 KB) than SF or AES. HIGHT works well in situations where strong diffusion and speed of execution are more important than compact memory usage.

TripleDES demonstrated CPU usage of 45–49 percent and encryption times of approximately 11.6 ms. According to these metrics, TripleDES is less effective than AES in terms of speed and resource usage, which makes it less suitable for contemporary cloud workloads even though it still operates fairly well.

With encryption and decryption times of 20.5–24.2 ms and 30.5–38.1 ms, respectively, RSA was the slowest algorithm tested. The cost of asymmetric operations in virtualized environments was reflected in its highest CPU usage, which peaked at over 70

RSA is still better suited for key exchange and authentication than for encrypting large amounts of data in cloud environments because of its high computational cost.

Due to its lightweight design, SEA was compared and reported:

Time of execution: 2.670 s

7.44 entropy

176 KB of RAM was used.

For minimal-use cases, SEA offers a respectable trade-off between performance and simplicity, despite not being the best in any one area.

According to the analysis, AES is the most effective and well-balanced cipher for cloud environments because it provides strong diffusion, low memory usage, and optimal speed. Notable lightweight choices that combine low resource requirements and good entropy are SF and HIGHT.

Algorithm	Dec Time (Brute Force)(hrs)	Avalanche (Key) (numeric)	Avalanche (Plain)(numeric)
DES	$2.0 * 10$ hrs	0.53	0.53
AES	$8.8 * 10^{22}$ hrs	0.5	0.55
Blowfish	$5.0 * 10^3$ hrs	0.29	0.18
Proposed	$3.2 * 10^{61}$ hrs	0.57	0.64

Fig. 14. Cloud Platform Performance Metrics - Avalanche Key

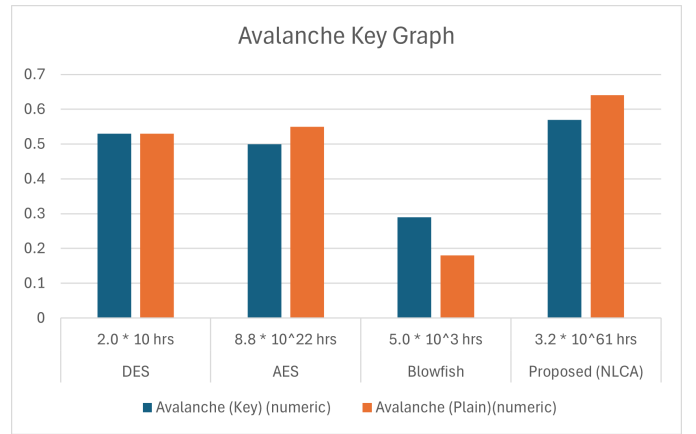


Fig. 15. Cloud Platform Performance Metrics -Avalanche Key Graph

RSA and TripleDES, on the other hand, have substantially greater computational costs, which makes them less appropriate for large-scale cloud operations. Despite their statistical interest, DES and SEA are not efficient enough for contemporary deployment.

These findings highlight the necessity of choosing algorithms for cloud-based systems that balance practical performance and security robustness. [38] [41] [42]

IV. DISCUSSION

The statistical results are interpreted platform-wise in this section, along with the findings' implications, study limitations, and suggested future research directions. It also provides answers to the three research questions that were introduced.

A. Interpretation

The statistical and thematic synthesis demonstrates that cryptographic performance is highly platform-dependent, with hardware architecture, OS-level optimizations, and application constraints all having an impact on suitability and efficiency.

IoT Platform

IoT Platform Statistics clearly show that in environments

with limitations, lightweight block ciphers perform better than conventional ciphers.

For Arduino-class boards, SPECK128 showed the best combination of speed (273 micro seconds), encryption throughput (824 Kbps), and low memory consumption (1164B RAM / 8848B ROM).

On TelosB motes, AES-CTR performed best in environments with OS-level optimizations, such as TinyOS, recording the lowest init (0.47 ms) and update (0.91 ms) times.

For large payloads, ChaCha20 performed better on the Raspberry Pi 3 than AES and DES.

Both XTEA and RECTANGLE scaled well for different payload sizes; RECTANGLE was better for large messages and XTEA for small ones.

The trade-off between cryptographic strength and responsiveness in real-time IoT use cases was confirmed by ASCON, which had the highest latency (849 microseconds) despite being secure.

Mobile Platform ChaCha20 demonstrated outstanding performance on both ARMv8-a (throughput of 138 MiB/s on Xiaomi) and ARMv7-a (throughput of 39 MiB/s on Samsung).

On low-end devices (Galaxy A20e), RC4 had the quickest execution time, demonstrating that legacy ciphers can still achieve low latency on low-power hardware.

With battery drain as low as 5.18 mAh, SPECK128 and LEA were the best options for passive background encryption.

Twofish was the top-performing block cipher on mid-range phones and demonstrated good memory efficiency.

When no hardware acceleration was available, AES-GCM performed worse than ChaCha20-Poly1305, confirming the importance of deploying ciphers with consideration for hardware.

Cloud Platform With encryption/decryption times as low as 3.1–3.5 ms, moderate CPU usage (41–47 percent), strong avalanche values (0.50 key, 0.55 plaintext), and low memory usage (14.7 KB), AES continued to provide the best overall performance.

Ideal for light-load cloud encryption, the SF cipher showed a respectable avalanche score (0.457), low RAM usage (22 KB), and high entropy (7.99).

Despite requiring more RAM (288 KB), HIGHT had the quickest execution time (1.615 s) and near-perfect entropy (7.9971) among lightweight ciphers.

The two algorithms with the highest computational costs were TripleDES and RSA. RSA's encryption times were over 24 ms, and its CPU usage was over 70 percent. This proved that asymmetric methods are not feasible for encrypting large amounts of data.

For low-priority data in memory-constrained cloud services, SEA provided simplicity (2.67 s exec, 7.44 entropy, 176 KB RAM).

*1) Answering the Research Questions: **RQ1. Which stream and block cipher algorithms are most frequently utilized on cloud, mobile, and Internet of Things platforms?***

It was discovered that a number of cryptographic algorithms

were commonly used in each platform type across the 29 papers that were chosen.

Block ciphers like AES, SPECK, PRESENT, XTEA, and RECTANGLE were frequently assessed in IoT environments. Additionally, stream ciphers such as ChaCha20, Trivium, and Grain were frequently tested, particularly in studies based on Raspberry Pi and Arduino.

While ChaCha20 and RC4 were the most commonly tested stream ciphers in the mobile domain, block ciphers like AES, Twofish, SPECK, and Blowfish were extensively researched.

AES was still the most popular cipher in the cloud platform, followed by Blowfish, ChaCha20, and older algorithms like 3DES. In specific cloud environments, lightweight substitutes such as NLCA, SF, and HIGHT also surfaced.

RQ2. How do these cryptographic methods stack up in terms of security strength and performance (speed, resource usage)? No single algorithm is superior in every way, according to the analysis.

In cloud and high-end mobile environments, AES continuously showed strong security features and outstanding performance, especially when hardware acceleration was available. However, in low-end IoT setups, its resource requirements were high.

On both cloud and mobile platforms, ChaCha20 combined low CPU usage, high security, and excellent throughput.

Although there have been some worries expressed regarding long-term cryptanalytic resistance, lightweight ciphers such as SPECK, XTEA, and RECTANGLE have proven effective for Internet of Things devices, providing low latency and memory usage.

With advantageous execution speeds and resource profiles, Twofish and Blowfish provided moderate-to-good security.

Due to known vulnerabilities, legacy ciphers like RC4 and 3DES are no longer advised, despite their strong speed performance.

RQ3. For each platform, which ciphers offer the optimal trade-off between efficiency and data security?

On IoT platforms, SPECK128 and AES-CTR provided the best trade-offs between speed, memory efficiency, and cryptographic strength. ChaCha20 also performed well on higher-capability IoT boards.

In mobile environments, ChaCha20-Poly1305 achieved the highest throughput with strong energy efficiency, while SPECK128, LEA, and Twofish offered reliable battery-conscious performance.

In cloud systems, AES was the most balanced in terms of security, execution speed, and memory usage. SF and HIGHT were strong lightweight alternatives for specific low-resource applications.

B. Implications

Hardware determines performance trends. ChaCha20, for instance, worked well on expensive smartphones but poorly on low-end IoT devices. Because of its acceleration, AES performed well in cloud and some mobile environments but poorly in energy-starved IoT nodes unless it was optimized.

The "best" cipher isn't universal. Rather, best-in-class ciphers differ in:

IoT: ChaCha20, AES-CTR, and SPECK128

Mobile: RC4, Twofish, SPECK128; ChaCha20

Cloud: SF, AES, and HIGHT

AES is still widely used in mobile and cloud computing (when hardware is supported), but it isn't always the best option for IoT unless it is optimized.

Particularly for mobile and IoT devices, lightweight cryptography is crucial. Instead of using standard block ciphers by default, the statistical results highlight the necessity of matching algorithm selection to performance constraints and application-specific requirements.

C. Limitations

There are several structural limitations to this statistical synthesis:

Lack of repeatable or standardized metrics (e.g., SPARKLE, Saturnin) led to the exclusion of many AEAD and post-quantum candidates.

Inconsistent benchmarking setups: The comparability of metrics may be impacted by variations in platform specifications, payload sizes, and operating system stacks among studies.

Limited coverage of asymmetry: RSA and other asymmetric algorithms were only evaluated for performance, not for use cases like key exchange, where they are usually implemented.

It was more difficult to create a single "efficiency score" for each cipher since energy and latency metrics were not always paired.

The chosen papers did not statistically measure the behavior of real-world application stacks, such as protocol overheads in TLS, MQTT, or cloud containerization. Most performance-focused studies did not address some performance metrics like energy, memory consumption, encryption and decryption speed.

D. Future Directions

Utilize standard testing configurations: To facilitate comparison of findings, future studies should test ciphers under identical hardware, file size, and condition conditions. Additionally, the statistical synthesis should be more detailed and more data set is needed for its future.

Do more extensive testing on lightweight ciphers: Additional performance testing is required for algorithms such as ASCON, TinyJambu, and others on platforms like cloud, mobile, and IoT.

Test ciphers in actual systems: Future research should evaluate ciphers' performance in complete systems, such as cloud storage or secure messaging apps, rather than just testing them. Make ciphers more platform-specific: Some ciphers are more effective when modified for specific devices. Future studies could examine how to optimize performance by adjusting ciphers for cloud servers, Raspberry Pis, or ARM chips.

Assist developers in selecting the appropriate cipher: Developing basic tools or guidelines to assist developers in selecting

the most appropriate cipher for their device and use case would be beneficial.

V. CONCLUSION

This study examined and contrasted block and stream ciphers on cloud, mobile, and Internet of Things platforms. It was discovered that no single cipher works best in every setting.

The most popular and well-balanced algorithm is still AES, particularly in hardware-supported cloud and mobile platforms.

On contemporary mobile and cloud systems, ChaCha20 provides exceptional speed and energy efficiency.

Because of their low memory and power consumption, lightweight ciphers like SPECK, XTEA, and others work best in Internet of Things settings.

This review offers a better understanding of which ciphers function best, where they are most effective, and why by comparing algorithms across real-world platforms. This will help inform more intelligent and useful decisions in future cryptographic design.

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